

Basic Principles of Medicine 2
Module: Endocrinology and Reproduction.
DLA Video

Infant Nutrition

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Recommended Reading

Lippincott Illustrated Reviews (**Lippincott**) 8th edition
Pages 441 and 450.

<https://meded.lwwhealthlibrary.com/book.aspx?bookid=3073>

- Information was also collected from publications that are not part of the reading assignment.
 - Human milk composition: Nutrients and bioactive factors
 - Olivia Ballard, *Pediatr. Clin. North. Am.* 2013, 60(1): 49-74
 - Ncbi.nlm.nih.gov/pmc/articles/PMC3586783/#R142
 - Review of infant feeding: Key features of breast milk and infant formula
 - CR Martin, *Nutrient* 2016, 8(5):279
 - Ncbi.nlm.nih.gov/pmc/articles/pMC48826921
 - Best baby-formula-like-breast-milk
 - Nursingmoms.net/ July 2020

Objectives: DLA: Nutritional Aspects of Lactation and Infant Nutrition

- DLA to be reviewed before “Nutritional Aspects of Pregnancy, Lactation and Infant Nutrition”

SOM.MK.III.BPM2.1.ER.2.BCHM.1055	Highlight the major differences in the composition of colostrum, transition milk and mature milk during lactation.
SOM.MK.III.BPM2.1.ER.2.BCHM.1056	Discuss the advantages of human milk compared to animal milk and infant formula.
SOM.MK.III.BPM2.1.ER.2.BCHM.1057	Review WHO guidelines about breastfeeding and HIV.
SOM.MK.III.BPM2.1.ER.2.BCHM.1058	Identify the age range of neonate and infant. Discuss Vitamin K deficiency bleeding (VKDB) in the neonate and infant.
SOM.MK.III.BPM2.1.ER.2.BCHM.1059	Identify micronutrients that are low in human breast milk. Highlight the dietary modifications (iron) to be implemented after 6 months of age.
SOM.MK.III.BPM2.1.ER.2.BCHM.1060	Describe bioavailability of iron and factors that reduce inorganic iron uptake.

Nutritional aspects of lactation and infant nutrition

A decorative graphic element consisting of a curved shape that starts as a thin light blue line on the left and curves downwards and to the right, filling with a teal color to become a solid shape at the bottom right.

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Human milk composition changes from colostrum, transition milk to mature milk



Colostrum has the main function to protect the newborn

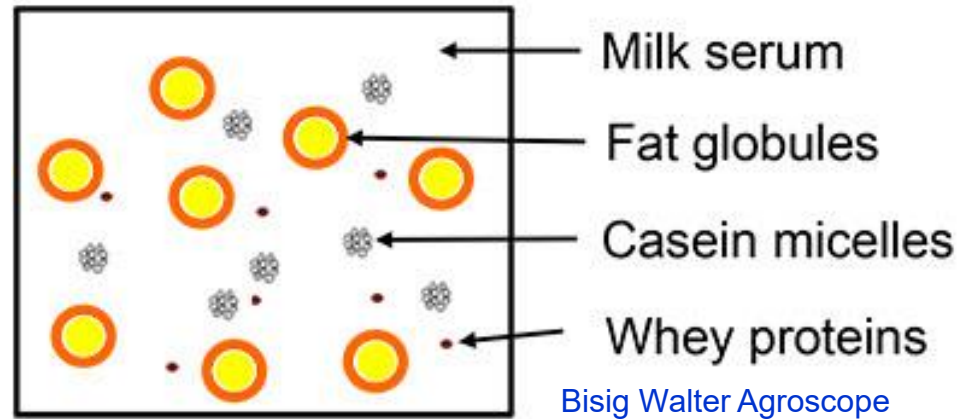
- **Colostrum** (first milk is the best for the newborn)
 - Produced during pregnancy and up to 4-5 days postpartum
 - **Primary functions are supporting the immune system and cellular growth**
 - Viscous fluid with **yellow** color due to high **vitamin A** content
 - Highest protein content mainly from secretory **IgA** and **lactoferrin**.
 - Lower in lactose than mature milk which has the functions to promote growth

Human Mature Milk Composition

(is dependent on nutrition and other factors)

Carbohydrate	Lactose and prebiotic HMO
Fat	TAGs and cholesterol Fatty acid composition in TAGs includes medium- and long chain fatty acids synthesized in the lactating mammary gland. TAGs are rich in oleic acid (36%) and contain PUFA ($\omega 6 > \omega 3$) from maternal diet including arachidonic acid (ARA) and DHA .
Protein	High whey to casein ratio in comparison to cow's milk Nutrients rich in essential and conditional essential amino acids Proteins with non-nutritive functions
Vitamins	Rich in vitamin A , B-vitamins and vitamin C Low in vitamin K and vitamin D, sunlight exposure of infant recommended

How can you describe milk?



Milk is a dispersion of

1. milk fat globules with TAGs
2. casein micelles with protein, calcium and phosphorus

all in a phase of milk serum with
lactose, whey protein vitamins and minerals.

Human milk contains two types of proteins: whey and casein.

Whey/casein ratio changes: highest in colostrum

Whey

- Mostly α -lactalbumin
- **Easier to digest than casein.** It does not precipitate at acid pH.
- Especially rich in **leucine** for stimulation of protein synthesis.

Casein

- **Casein micelles** contain water, **calcium** and **phosphorus**
- **Slow digestion** of the protein, precipitates at acid pH (curd)
- Especially rich in **proline** for collagen synthesis.

What is the advantage of whey in human milk?

- a. **α -lactalbumin:** is the **predominant protein** in **whey** and is often referred to as the **whey protein**. It is rich in cysteine and tryptophan.
- b. **Lactoferrin:** binds iron and **inhibits** growth of destructive **iron-dependent bacteria** in the **GI tract** of neonates and infants.
- c. **Lysozyme:** is an **enzyme** that **protects** the infant against **E. coli** and **Salmonella**. It **promotes** the growth of **healthy** intestinal **flora** and has **anti-inflammatory functions**.
- d. **Immunoglobulins:** mostly **IgA** which inhibits binding of pathogens, and **protects the immature intestine of babies**.

http://ansci.illinois.edu/static/ansc438/Milkcompsynth/milkcom_protein.html modified

Human Milk compared to Cow milk and Infant Formula

Breast Milk vs Cow and Infant Formula Composition					
Content per liter	Human	Cow	Infant Formulas		
			Preterm	Conv.	Milk-free
Energy (kcal)	690	660	676	676	676
Protein (g)	9	35	20	18	17
Taurine (mg)	80-40	1	48-57	40	
Fat (g)	45	37	35	35	35
Carbohydrates	68	49	74	79	75
Lactose	68	49	30	55	0

Whey/Casein ratio of 80/20 – 60/40

Whey/Casein ratio of 20/80

DIFFERENCES BETWEEN COW MILK, FORMULA MILK AND BREAST MILK

	BREAST MILK	COW MILK	FORMULA MILK
WATER	Enough	Extra needed	May need extra
ENERGY	Equal	Equal	Equal
PROTEIN	Correct amount easy to digest	Too much difficult to digest 	Partially corrected
CARBOHYDRATE	Lactose – plenty oligosaccharides	Lactose – less oligosaccharides	Lactose + Sucrose Lacks lipase
FAT	EFA's present Lipase to digest	No EFA's No lipase	Some EFA's added No lipase
VITAMINS	Adequate depending on mom's nutritional adequacy	Low vitamin A, C and Iron	Vitamin / mineral added – usually enough
ANTI-INFECTIVE FACTORS	IgA, Lactoferrin, Lysozyme, etc	None	None
GROWTH FACTORS	Present	None	none

Mainly 3 types of infant formulas

1. **Cow milk-based** formula (often whey added)
2. **Soy-based** formula (not for premature infants)
3. **Specialized formula** (hypoallergenic or lactose-free)

Additives may include:

Lactoferrin, HMO, ARA, DHA, lutein, taurine, choline, vitamins, and many other components dependent on the chosen formula.

More new information July 2020 at nursingmoms.net/best-baby-formula-like-breast-milk

HIV and breastfeeding 2020 (cdc.gov/breastfeeding) citation:



- “The **CDC** and **American Academy of Pediatrics** recommend that in the **US**, where mothers have access to clean water and affordable infant formula, HIV-infected mothers **completely avoid** breastfeeding their infants, regardless of anti-retroviral therapy (ART) and maternal viral load.”
- “In **resource-limited settings** like in some parts of **Afrika**, the **WHO** recommends that HIV- infected mothers **breastfeed exclusively** for the first 6 months of life, and continue breastfeeding for at least 12 months, with addition of supplementation of complementary foods. These mothers should be given **ART** to reduce the risk of transmission through breastfeeding.”
- **WHO:** “**Breastfeeding is one of the foundations of child health**, development and survival, especially when diarrhea, pneumonia and undernutrition are common causes of mortality among children younger than five years.”

Now let's look at the gorgeous Infants:

Age range of neonate and infant



Neonatal is up
to 28 days old



Infants (babies)
are up to 1-year-old

Neonatal hemorrhagic disease

Vitamin K Deficiency Bleeding (VKDB) can occur during the **first week** to a **small number** of neonates (0.3 to 1.7 %) who do not **receive vitamin K treatment**. In most cases it is not lethal.

VKDB is more severe when it happens later in infants up to **6 months** of age and then it may be **lethal**.

Babies can be perfectly **healthy** one day then suddenly become **neurologically ill** the **next day**. The cause of **death** is either **cerebral** or **intestinal hemorrhage**.

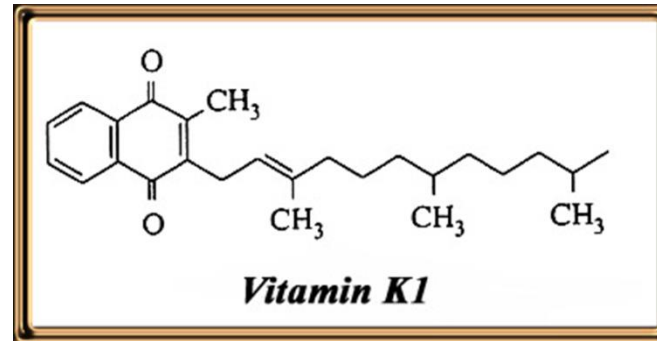
The American Academy of Pediatrics has recommended vitamin K1 injections for newborns in order to prevent VKDB



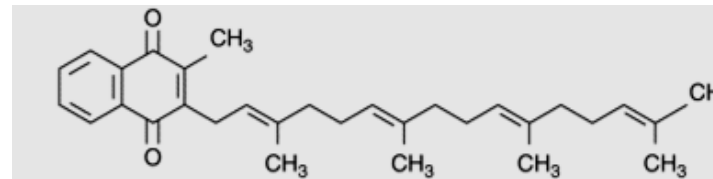
1. **Vitamin K** is mostly **injected** (1 mg) but concerns are raised.
2. **Vitamin K** can also be given **orally** which is more complicated:
 - a. It has to be given in **multiple doses** (parents may forget).
 - b. Full dose might not be **absorbed** (baby spits it out).
 - c. Has to be given **after food intake** for better absorption.

Vitamin K is a lipid-soluble vitamin that is needed for efficient blood clotting

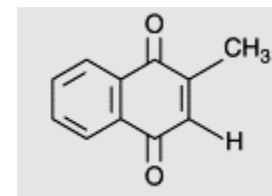
- From plants:
Phylloquinone K1



- From bacteria:
Menaquinone K2



- Drug: Menadone K3



Why are neonates vitamin K deficient?

1. **Vitamin K2 is lacking** in the newborns **sterile intestine (no bacteria)**.



2. **Human milk is very low in vitamin K.**
Formulas are now vitamin K supplemented.



3. **Vitamin K1 is mainly found in green vegetables** which are not part of the **neonatal diet**.



Dietary modification for infants older than 6 months:



Supplementation with
mashed solid foods
in addition to milk or formula.



food	6 months	7–9 months	8-12 months
Breast Milk or Formula	4–5 feedings	3–5 feedings	3–4 feedings
Cereal/ Grain	2 meals	2 meals	2 meals
Fruits	2 meals	2 meals	2 meals
Vegetables	2 meals	2 meals	2 meals
Fruit Juice (optional)		1 serving	1 serving
Protein Foods		1 meal	2 meals

* A meal is approximately 2 to 4 tbsp

SOM.MK.III.BPM2.1.ER.2.BCHM.1059

Some pureed food rich in vitamins and minerals

- **Avocado:**
 - Rich in essential fatty acids and minerals.
- **Banana:**
 - Electrolytes and calcium.
- **Carrots, peas, green beans, broccoli:**
 - β -carotene and minerals.
- **Squashes, sweet potato:**
 - Easy to digest carbohydrates.
- **Pears:**
 - Least acid fruit, Vit K and copper.



Infant nutrition has to provide **iron** after 6 months of age

- **During pregnancy** the mother provides iron for an increased **iron depot** to the **fetal liver** which will supply the **baby** up to **six months** after birth.
- **Dietary iron uptake** is **needed** in the **6-month-old baby**.
- **Human milk** is **low** in **iron** (0.35 mg/L) and **infants** need about **11 mg** of **iron** per day which surely **cannot** be provided by **milk alone**.

Which food for infants is rich in iron?

Spinach puree is often given to infants because it is **rich in iron but the bioavailability is low.**

Inorganic iron has to be in the **ferrous state** for **uptake into intestinal mucosal cells.**

In addition, the **phytates** and **polyphenols** in spinach **bind iron** and reduce the non-heme iron **uptake.**

Meat puree is often given to infants because it contains **heme-iron** which is **better absorbed.**



**Thank you,
please email ASKBCHM@sgu.edu
if you have questions.**